

Readme

База складається з 7 таблиць. create_tables.sql

Зв'язки:

One-to-One: Таблиці patients та patient_profiles.

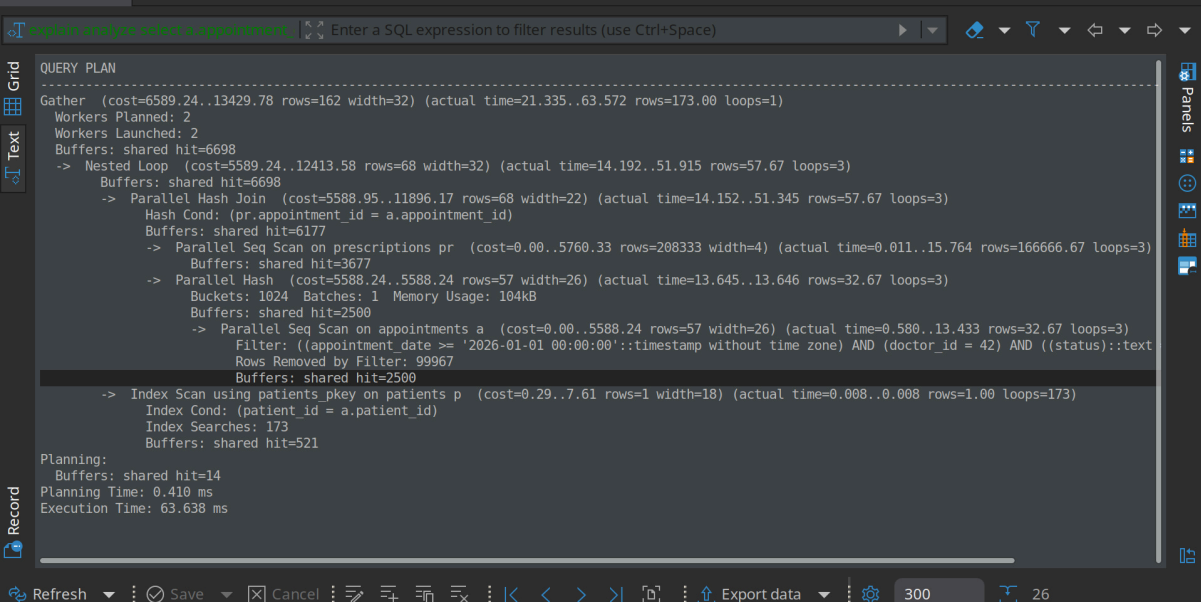
One-to-Many: departments і doctors, patients і appointments.

Many-to-Many: appointments та medications.

Оптимізація(Tests_explain_analyze.sql): Додано індекси для foreign keys та інших колонок.

(additional_points)Створено ролі з різними доступами. Два view, перший приховує чутливу інформацію, другий створює зручний перегляд запланованих прийомів. Процедура забезпечує зміну статусу візиту з перевітками, а функція спрощує створення нових візитів.

Тести та порівняння до та після додання індексів:



```
explain analyze select a.appointment_id
QUERY PLAN
Gather  (cost=6589.24..13429.78 rows=162 width=32) (actual time=21.335..63.572 rows=173.00 loops=1)
  Workers Planned: 2
  Workers Launched: 2
  Buffers: shared hit=6698
  -> Nested Loop  (cost=5589.24..12413.58 rows=68 width=32) (actual time=14.192..51.915 rows=57.67 loops=3)
    Buffers: shared hit=6698
    -> Parallel Hash Join  (cost=5588.95..11896.17 rows=68 width=22) (actual time=14.152..51.345 rows=57.67 loops=3)
      Hash Cond: (pr.appointment_id = a.appointment_id)
      Buffers: shared hit=6177
      -> Parallel Seq Scan on prescriptions pr  (cost=0.00..5760.33 rows=208333 width=4) (actual time=0.011..15.764 rows=166666.67 loops=3)
        Buffers: shared hit=3677
      -> Parallel Hash  (cost=5588.24..5588.24 rows=57 width=26) (actual time=13.645..13.646 rows=32.67 loops=3)
        Buckets: 1024 Batches: 1 Memory Usage: 104kB
        Buffers: shared hit=2500
        -> Parallel Seq Scan on appointments a  (cost=0.00..5588.24 rows=57 width=26) (actual time=0.580..13.433 rows=32.67 loops=3)
          Filter: ((appointment_date >= '2026-01-01 00:00:00'::timestamp without time zone) AND (doctor_id = 42) AND ((status)::text
          Rows Removed by Filter: 99967
          Buffers: shared hit=2500
        -> Index Scan using patients pkey on patients p  (cost=0.29..7.61 rows=1 width=18) (actual time=0.008..0.008 rows=1.00 loops=173)
          Index Cond: (patient_id = a.patient_id)
          Index Searches: 173
          Buffers: shared hit=521
    Planning:
      Buffers: shared hit=14
      Planning Time: 0.410 ms
      Execution Time: 63.638 ms
```

Results 1 x

explain analyze select a.appointment_date, Enter a SQL expression to filter results (use Ctrl+Space)

```

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Nested Loop (cost=9.64..2574.03 rows=162 width=32) (actual time=1.307..14.633 rows=173.00 loops=1)
  Buffers: shared hit=1021 read=103
  -> Nested Loop (cost=9.22..2153.05 rows=97 width=36) (actual time=1.157..11.835 rows=98.00 loops=1)
    Buffers: shared hit=825 read=4
    -> Bitmap Heap Scan on appointments a (cost=8.92..1414.98 rows=97 width=26) (actual time=0.342..1.460 rows=98.00 loops=1)
      Recheck Cond: (doctor_id = 42)
      Filter: ((appointment_date >= '2026-01-01 00:00:00'::timestamp without time zone) AND ((status)::text = 'completed'::text))
      Rows Removed by Filter: 496
      Heap Blocks: exact=531
      Buffers: shared hit=531 read=4
      -> Bitmap Index Scan on idx_appointments_doctor_id (cost=0.00..8.90 rows=597 width=0) (actual time=0.224..0.225 rows=594.00 loops=1)
        Index Cond: (doctor_id = 42)
        Index Searches: 1
        Buffers: shared read=4
      -> Index Scan using patients_pkey on patients p (cost=0.29..7.61 rows=1 width=18) (actual time=0.104..0.104 rows=1.00 loops=98)
        Index Cond: (patient_id = a.patient_id)
        Index Searches: 98
        Buffers: shared hit=294
    -> Index Only Scan using idx_prescriptions_appointment_id on prescriptions pr (cost=0.42..4.31 rows=3 width=4) (actual time=0.026..0.027 rows=1.7)
      Index Cond: (appointment_id = a.appointment_id)
      Heap Fetches: 0
      Index Searches: 98
      Buffers: shared hit=196 read=99
Planning:
  Buffers: shared hit=120 read=16
  Planning Time: 3.114 ms
  Execution Time: 14.737 ms

```

Refresh Save Cancel

Export data 300 27

ERD cxemi:

